Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

1. TEAM AND PROJECT OVERVIEW

Team Name

Kopi-Paste Coders

Selected Domain

Domain 2 - ALPHA RED

Selected SDG Track (1–17)

SDG 8 and SDG 12

Project / Solution Name

ESG-Passport

Name (Leader)

Lee Boon Yew

Email & Phone (Leader)

leeboonyew6868@gmail.com | +60107659969

University Affiliation(s)

Tunku Abdul Rahman University of Management and Technology (TAR UMT)

Number of Team Members

4

2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.1. The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Malaysian micro and small businesses supplying larger companies face a recurring administrative burden: responding to environmental, social and governance (ESG) questionnaires using records scattered across paper receipts, utility bills, spreadsheets and messages. Owners and administrative staff must interpret unfamiliar questions, locate supporting documents and reconcile quantities, units and reporting periods, often without dedicated sustainability expertise.

Malaysia's National Sustainability Reporting Framework (NSRF) strengthens the relevance of this challenge. It adopts IFRS S1 and IFRS S2 as baseline sustainability disclosure standards, with phased implementation beginning in 2025. Although ordinary SMEs are not automatically subject to these reporting requirements, those within larger companies' value chains may face requests for sustainability data, including information supporting Scope 3 emissions reporting.

The practical gap is between everyday business records and credible, reusable disclosures. Electricity expenditure does not directly establish energy consumption or emissions, while unsupported staff estimates cannot provide the same evidence as documented measurements. Repeated manual interpretation can introduce errors, obscure missing information and delay customer responses. These difficulties may weaken suppliers' ability to demonstrate readiness for procurement requirements and identify opportunities to reduce resource use.

ESG Passport addresses this specific burden through a proposed Orbyte-compatible preparation workflow: organising evidence, mapping questionnaire requirements to the Simplified ESG Disclosure Guide (SEDG), identifying gaps and retaining human review and approval. SEDG provides an SME-oriented reference for organising disclosures. The competition prototype uses team-created synthetic records and prepares reviewable outputs without submitting statutory filings.

2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.2. The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

ESG Passport connects document intake, SEDG questionnaire mapping, evidence matching and owner-approved sharing in one monthly SME workflow.

Users upload fictional electricity, water, fuel or waste records. When explicitly enabled, Gemini transcribes document text; Grafilab is an optional alternative or fallback. A Python parser converts the text into editable JSON candidates. RapidOCR supplies an offline fallback. Original files and original extraction results are retained, and quantities enter calculations only after a person confirms them. Browser-only OCR is a separate local scanning option.

Questionnaire mapping uses deterministic English/Bahasa Malaysia rules to suggest five Basic environmental SEDG codes. Older rules suggest selected Social/Governance codes. Users review the proposals; this mapping currently makes no LLM call. Evidence checks connect reviewed requirements to confirmed activities and flag missing periods, source corroboration and business boundaries.

Versioned calculation rules perform unit conversions and configured emissions calculations. The existing profile remains separate from the stricter preparation checks. Before sharing, the simulated owner reviews selected evidence and explicitly acknowledges outstanding gaps. An immutable snapshot and activity log record the decision.

The implementation uses Next.js/React, TypeScript, FastAPI/Python, PostgreSQL and private Supabase Storage. Hosting on organiser-provided Replit Core supports frontend and backend deployment. These apply GenAI API, database and cloud-deployment skills from Phase 1. All four team members completed the mandatory Phase 1 workshops. Grafilab API usage is evidenced by saved comparison responses. Workshop/sponsor usage proof remains to be added.

The competition demo uses team-created synthetic records. Planned improvements include exposing requirement highlights and priority queues, unifying the OCR adapters and measuring complete manual-versus-assisted preparation time.

3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.1. Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Frontend:	Next.js 16, React 19, TypeScript, Tailwind CSS.
Backend:	Python 3.12, FastAPI, SQLAlchemy, Pydantic, HTTPX.
Data:	PostgreSQL and private Supabase Storage.
OCR:	Gemini / optional Grafilab Qwen; RapidOCR CPU and PDF text extraction as offline alternatives. Tesseract.js/PDF.js support a separate browser-only scanner.
SEDG:	shared deterministic EN/BM mapping, evidence/coverage helpers, versioned unit conversions and explicit owner approval. No live Orbyte connector.
Hosting:	Organiser-provided Replit Core hosts the frontend and backend.
Sponsor usage:	Replit Core for hosting and deployment.

System architecture and workflow

SME clerks and owners use the Next.js/React web interface to upload monthly evidence, review OCR suggestions and prepare customer ESG responses. The interface communicates with the FastAPI/Python backend, which coordinates validation, questionnaire matching, evidence checks and calculations.

Gemini or optional Grafilab can transcribe documents; RapidOCR is an offline CPU alternative. Supabase PostgreSQL stores reviewed records, while private Supabase Storage retains original documents. Deterministic English/Bahasa Malaysia rules suggest supported SEDG codes, and versioned calculation rules flag missing or inconsistent data.

People must confirm extracted quantities before calculations. The simulated owner reviews disclosures, acknowledges outstanding gaps and approves selected evidence for sharing. The synthetic demonstration has no live Orbyte connector or statutory filing integration.

3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.2. Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

We defined one task around an SME owner and administrator: prepare a customer ESG response from monthly records without losing evidence or control. These are prototype personas, not claims of completed customer interviews.

We prototyped document review, code confirmation, gap acknowledgement and owner approval around that task. Synthetic Malaysian-style utility bills, receipts and weighbridge records let us exercise alternative layouts and ambiguous fields without using real customer records in the demo.

Technical evaluation separated transcription, structured field accuracy, latency and evidence coverage. An eight-image paired development comparison informed the operational preference for Gemini. Twelve offline backend/AI bridge tests passed on 24 October; full backend and deployment acceptance checks remain outstanding in this local environment.

For Phase 2, we will observe representative users completing matched manual and assisted tasks, including corrections and approvals. We will fix the known OCR timeout, estimated-reading and receipt-date issues, then verify hosted configuration, recoverability and judge access. The current deployment design uses a shared synthetic workspace with simulated roles; real customer use requires stronger identity, tenant isolation and data-retention controls. Orbyte integration is an interface assumption, not a live connector.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess Innovation and Creativity as well as Feasibility of build. Vague or unsupported projections may weaken your case for advancing to Phase 2.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Primary: SDG 8.2 - Productivity through technological upgrading and AI-assisted ESG preparation. SDG 8.3 - SME growth by lowering the administrative burden of responding to customer ESG requests. Enabling: SDG 8.4 - Resource efficiency through improved visibility of energy, water and waste data; savings not yet verified. SDG 12.6 - Reporting readiness through documented evidence and owner-reviewed sustainability disclosures. UN sources: https://sdgs.un.org/goals/goal8 https://sdgs.un.org/goals/goal12
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Survey evidence: 37% of ESG adopters surveyed were confused about standards/frameworks; 51% cited market demand. (https://www.alliancebank.com.my/About-Us/Media-Centre/ESG-Awareness-Among-SMEs-Surges-Sixfold-Alliance-Bank-ESG-2-0-Report) Project pilot baseline: 167.72 s manual time per task (one tester; 10 paired synthetic trials). Each task block included four synthetic documents, five questions and 40 scored fields. This is a process-time baseline, not measured resource savings.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Metric Category	Your Response
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided*	Primary KPI: Percentage reduction in document-processing completion time. Formula: (Manual total time - AI-assisted total time) / Manual total time x 100. Observed: 1,677.19 s manual vs 866.08 s AI-assisted = 48.4% less total time (10 paired trials; 167.72 vs 86.61 s per task). Assistance used CPU RapidOCR and deterministic rules. This internal pilot involved one tester and reused materials; it excluded the complete hosted approval-and-sharing journey. Secondary KPI: Field-level accuracy; manual 94.5% vs assisted 98.0% (+3.5 percentage points).
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Phase 2 target: At least 30% lower median preparation time against matched manual work, without reducing reviewed field-level accuracy. Validate using new, unseen synthetic records and an approval step. The observed 48.4% pilot reduction in total time is preliminary (one tester, n=10); it does not demonstrate reduced resource use or median time savings. SDG link: 8.2 productivity; 8.3 SME growth; 8.4 resource-use visibility; 12.6 reporting readiness.

5. ELIGIBILITY & READINESS CHECKLIST

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & ORIGINALITY DECLARATION

6.1. Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6. IP & ORIGINALITY DECLARATION

6.2. Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Next.js / React / FastAPI	Application code frameworks	MIT; versions in repository
RapidOCR / Tesseract / ONNX	Pre-trained OCR + runtimes	Apache-2.0 / MIT; weight terms separately
Gemini / Grafilab Qwen	Hosted pre-trained API models	Provider API terms; not Apache relicensed
Team synthetic records	360 images + paired labels	Apache-2.0 original contributions

6. IP & ORIGINALITY DECLARATION

6.3. Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. DECLARATION

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature]



[Date]

25 October 2026

Team leader signature and actual signing date required.